4.1.2 Conservation and dissipation of energy

4.1.2.1 Energy transfers in a system

Content	Key opportunities for skills development
Energy can be transferred usefully, stored or dissipated, but cannot be created or destroyed. Students should be able to describe with examples where there are energy transfers in a closed system, that there is no net change to the total energy. Students should be able to describe, with examples, how in all system changes energy is dissipated, so that it is stored in less useful ways. This energy is often described as being 'wasted'.	

Content	Key opportunities for skills development
Students should be able to explain ways of reducing unwanted energy transfers, for example through lubrication and the use of thermal insulation. The higher the thermal conductivity of a material the higher the rate of energy transfer by conduction across the material. Students should be able to describe how the rate of cooling of a building is affected by the thickness and thermal conductivity of its walls. Students do not need to know the definition of thermal conductivity.	WS 1.4 AT 1, 5 Investigate thermal conductivity using rods of different materials.

Required practical activity 2 (physics only): investigate the effectiveness of different materials as thermal insulators and the factors that may affect the thermal insulation properties of a material.

AT skills covered by this practical activity: AT 1 and 5.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in Key opportunities for skills development (page 90).

4.1.2.2 Efficiency

Content	Key opportunities for skills development
The energy efficiency for any energy transfer can be calculated using the equation: $efficiency = \frac{useful\ output\ energy\ transfer}{total\ input\ energy\ transfer}$ Efficiency may also be calculated using the equation: $efficiency = \frac{useful\ power\ output}{total\ power\ input}$	MS 3b, c Students should be able to recall and apply both equations. MS 1a, c, 3b, c Students may be required to calculate or use efficiency values as a decimal or as a percentage.
(HT only) Students should be able to describe ways to increase the efficiency of an intended energy transfer.	(HT only) WS 1.4